ECE 340 **Fall 2026**

Homework Assignment #4

DUE: FRIDAY, SEPTEMBER 25, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME BEFORE 5:00PM ON THE DUE DATE.

1. DOPING AND MOBILITY

(A). Show that the minimum conductivity of a semiconductor sample occurs when $n_0 = n_i\sqrt{\frac{\mu_p}{\mu_n}}$ and give the expression for σ_{min}.

(B). Suppose a sample of silicon is uniformly doped with $1 \times 10^{17}\ cm^{-3}$ phosphorus atoms. If $3 \times 10^{17}\ cm^{-3}$ boron atoms are then added to the sample, by approximately what factor has the resistivity changed (i.e. what is $\frac{\rho_{new}}{\rho_{old}}$)? What is the conductivity type? (Reference Fig. 3-23 from the textbook.)

(C). In many situations, higher conductivity is preferred. Which doping type should be used to achieve this whenever possible? **Solution** **20 points in total**

(A). **8 points**

The conductivity of a semiconductor is

$$\begin{aligned}\sigma &= q(n_0\mu_n + p_0\mu_p)\\ &= q\left(n_0\mu_n + \frac{n_i^2}{n_0}\mu_p\right).\end{aligned}$$

To find the minimum conductivity, differentiate with respect to n_0 and set the result equal to zero:

$$\begin{aligned}\frac{d\sigma}{dn_0} &= q\left(\mu_n - \frac{n_i^2}{n_0^2}\mu_p\right) = 0\\ \implies n_0^2 &= n_i^2\frac{\mu_p}{\mu_n}\\ \implies n_0 &= n_i\sqrt{\frac{\mu_p}{\mu_n}}.\end{aligned}$$

Substituting this concentration into the conductivity equation,

$$\sigma_{min} = q\left(n_i\sqrt{\frac{\mu_p}{\mu_n}}\mu_n + n_i\sqrt{\frac{\mu_n}{\mu_p}}\mu_p\right)$$
$$= 2qn_i\sqrt{\mu_n\mu_p}.$$

(B). **8 points**

Initially, the sample is n-type, with an electron concentration of

$$n_{0,old} \approx N_D = 1\times10^{17}cm^{-3}.$$

After adding boron, the acceptors compensate the donors, giving

$$p_{0,new} \approx N_A - N_D$$
$$= 3\times10^{17} - 1\times10^{17}$$
$$= 2\times10^{17}cm^{-3}.$$

The new conductivity type is **p-type** because the acceptor concentration exceeds the donor concentration.

Using the resistivity relationship,

$$\rho = \frac{1}{\sigma}$$
$$\approx \frac{1}{qn\mu_n} \quad \text{(n-type)},$$
$$\approx \frac{1}{qp\mu_p} \quad \text{(p-type)}.$$

Therefore,

$$\frac{\rho_{new}}{\rho_{old}} = \frac{n_{0,old}\mu_{n,old}}{p_{0,new}\mu_{p,new}}$$
$$= \frac{(1\times10^{17})(800)}{(2\times10^{17})(200)}$$
$$= 2.$$

The resistivity increases by approximately a factor of **2**.

The numerical answer depends on the mobilities estimated from Fig. 3-23. Acceptable values may range from approximately 1.5 to 2.5 if the correct method is used.

(C). **4 points**

For comparable majority-carrier concentrations,

$$\sigma_n \approx qn\mu_n,$$
$$\sigma_p \approx qp\mu_p.$$

From Fig. 3-23, electron mobility is generally higher than hole mobility in silicon.

Therefore, **n-type doping** is preferred when higher conductivity is desired, assuming comparable carrier concentrations and doping conditions.

2. Recombination and Generation

Consider p-type silicon at $T = 300$ K with an acceptor doping concentration of $N_A = 3 \times 10^{16}\ cm^{-3}$. The excess carrier lifetime is $\tau_n = \tau_p = 1\ \mu$s.

(A). Determine the thermal equilibrium recombination rate of electrons. Is the thermal equilibrium recombination rate of holes equal to that of electrons? Briefly explain your answer.

(B). Under continuous optical illumination, the steady-state excess carrier concentrations are $\delta n = \delta p = 5 \times 10^{14}\ cm^{-3}$. Determine whether the semiconductor is under low-level or high-level injection. Briefly justify your answer.

(C). Calculate the optical generation rate g_{op} required to maintain these excess carrier concentrations at steady state. **Solution** **20 points in total**

(A). **8 points**

$$\begin{aligned} p_0 &\approx N_A = 3 \times 10^{16} cm^{-3} \\ n_0 &= \frac{n_i^2}{p_0} \\ &= \frac{(1.5 \times 10^{10})^2}{3 \times 10^{16}} \\ &= 7.5 \times 10^3 cm^{-3}. \end{aligned}$$

For the thermal equilibrium recombination rate of electrons,

$$\begin{aligned} R_{n0} &= \frac{n_0}{\tau_n} \\ &= \frac{7.5\times10^3}{1\times10^{-6}} \\ &= 7.5\times10^9 cm^{-3}s^{-1}. \end{aligned}$$

The thermal equilibrium recombination rates of electrons and holes are **equal** because each recombination event removes one electron and one hole.

(B). **4 points**

Since $\delta p \ll p_0$, the semiconductor is under **low-level injection**.

(C). **8 points**

Under steady-state illumination, the optical generation rate is proportional to the amount of excess carriers:

$$\begin{aligned} g_{op} &= \frac{\delta n}{\tau_n} \\ &= \frac{5\times10^{14}}{1\times10^{-6}} \\ &= 5\times10^{20} cm^{-3}s^{-1}. \end{aligned}$$

3. Fermi Level Invariance

(A). Consider a doped Si sample (sample 1) where $E_{F_1} - E_i = 0.3eV$. Consider another doped Si sample (sample 2) where $E_{F_2} - E_i = -0.3eV$. Draw the individual band diagrams at 300K for samples 1 and 2 assuming they are not touching. Label E_c, E_v, E_{F_1}, E_{F-2}, E_i, and the distances between each. Assume the bandgap of Si to be 1.1 eV.

(B). Samples 1 and 2 from part (A) are brought together such that a perfect junction forms between them. Ignoring the shape of E_c, E_v, and E_i near the junction, draw the band diagram at equilibrium paying attention to E_{F_1} and E_{F_2} through the entire structure and E_c, E_v, and E_i away from the junction.

(C). Based on the diagram from part (B), if an electron in sample 1 had the same total energy as an electron in the conduction band edge of sample 2, how much kinetic energy would it have?

Solution **20 points in total**

(A). **8 points levels.**



Figure 3.1: Answer to Question 3 (A).

(B). **8 points**

Answer on Figure 3.2 (next page)

(C). **4 points**

$$\begin{aligned} KE &= (E_{c2} - E_{F2}) - (E_{c1} - E_{F1}) \\ &= (0.55 + 0.3) - 0.25 \\ &= 0.6eV \end{aligned} \tag{3.1}$$

Note: This is the same as the total band offset (the built-in potential)

4. Optical Absorption and Photoconductivity

(A). Use Fig. 4.1 and Appendix III (Fig. 4.2) to find the following for GaAs and CdS: (i) the wavelength corresponding to their bandgap energy (E_g), and (ii) the wavelength corresponding to an absorption coefficient $\alpha = 1000\ cm^{-1}$.

(B). A Si sample of 4 μm thickness is illuminated with light at a wavelength of 800 nm and an optical power of 1 mW. What is the absorption coefficient of Si in this case? How many electron-hole pairs (EHPs) are created in this sample per second (assuming 100% quantum efficiency and that 30% of the incident light is reflected)?

(C). Assume the length of the Si sample in part (B) is 500 μm, the width is 250 μm, and the thickness is still 4 μm. Assuming that the generated EHPs in (B) are evenly distributed in the sample, calculate the optical generation rate, g_{op}, defined as the **number of generated EHPs in the sample per second per unit volume**.

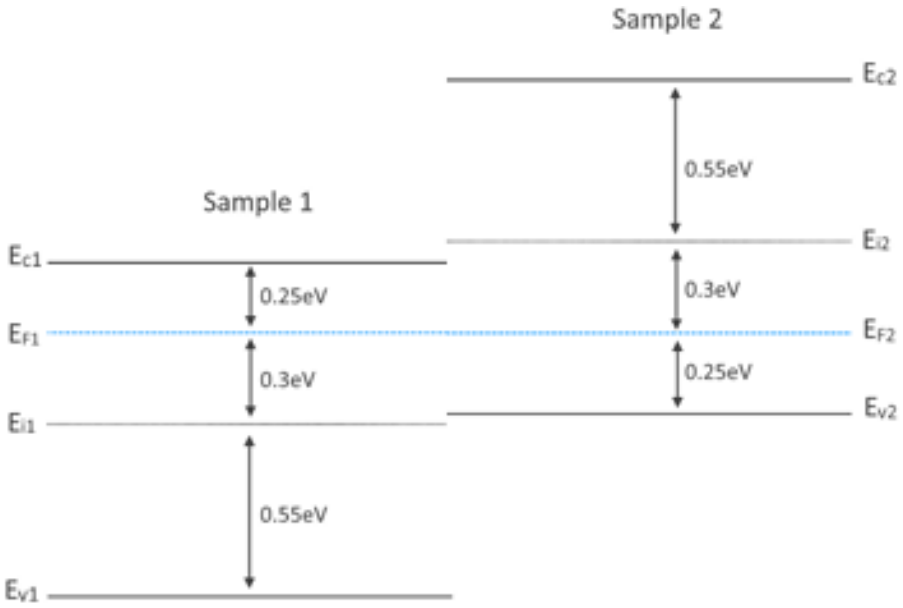


Figure 3.2: Answer to Question 3 (B).

Solution **20 points in total**

(A). **6 points in total.**
For GaAs, the bandgap energy is

$$E_g = 1.43eV.$$

The corresponding wavelength is

$$\begin{aligned}\lambda &= \frac{hc}{E_g} \\ &= \frac{1240eV \cdot nm}{1.43eV} \\ &= 867nm.\end{aligned}$$

From Fig. 4.1, the wavelength corresponding to an absorption coefficient of $\alpha = 1000\ cm^{-1}$ is

$$\lambda_{\alpha=1000cm^{-1}} = 860nm.$$

For CdS, the bandgap energy is

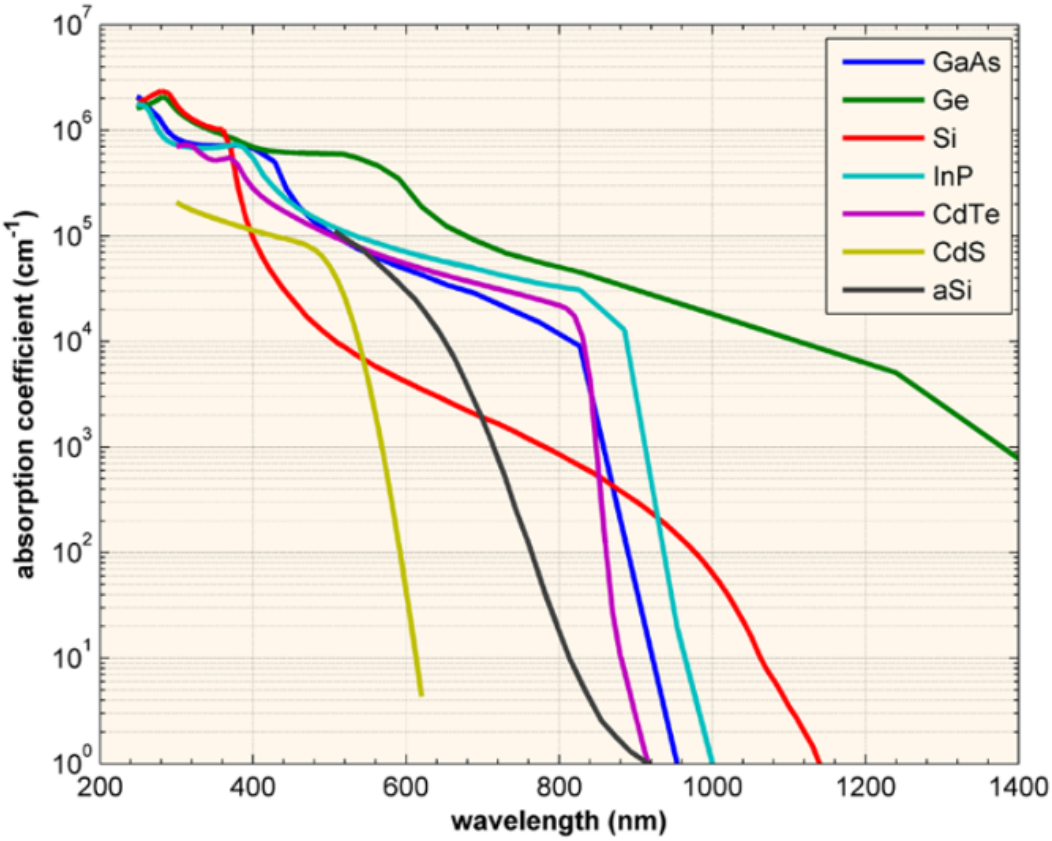


Figure 4.1: Absorption coefficient of various semiconductors

		E_g (eV)	μ_n (cm^2/V-s)	μ_p (cm^2/V-s)	m_n^*/m_o (m_l, m_t)	m_p^*/m_o (m_{lh}, m_{hh})	a (Å)	ϵ_r	Density (g/cm^3)	Melting point (°C)
Si	(*i*/*D*)	1.11	1350	480	0.98, 0.19	0.16, 0.49	5.43	11.8	2.33	1415
Ge	(*i*/*D*)	0.67	3900	1900	1.64, 0.082	0.04, 0.28	5.65	16	5.32	936
SiC (α)	(*i*/*W*)	2.86	500	—	0.6	1.0	3.08	10.2	3.21	2830
AlP	(*i*/*Z*)	2.45	80	—	—	0.2, 0.63	5.46	9.8	2.40	2000
AlAs	(*i*/*Z*)	2.16	1200	420	2.0	0.15, 0.76	5.66	10.9	3.60	1740
AlSb	(*i*/*Z*)	1.6	200	300	0.12	0.98	6.14	11.0	4.26	1080
GaP	(*i*/*Z*)	2.26	300	150	1.12, 0.22	0.14, 0.79	5.45	11.1	4.13	1467
GaAs	(*d*/*Z*)	1.43	8500	400	0.067	0.074, 0.50	5.65	13.2	5.31	1238
GaN	(*d*/*Z*, *W*)	3.4	380	—	0.19	0.60	4.5	12.2	6.1	2530
GaSb	(*d*/*Z*)	0.7	5000	1000	0.042	0.06, 0.23	6.09	15.7	5.61	712
InP	(*d*/*Z*)	1.35	4000	100	0.077	0.089, 0.85	5.87	12.4	4.79	1070
InAs	(*d*/*Z*)	0.36	22600	200	0.023	0.025, 0.41	6.06	14.6	5.67	943
InSb	(*d*/*Z*)	0.18	10^5	1700	0.014	0.015, 0.40	6.48	17.7	5.78	525
ZnS	(*d*/*Z*, *W*)	3.6	180	10	0.28	—	5.409	8.9	4.09	1650*
ZnSe	(*d*/*Z*)	2.7	600	28	0.14	0.60	5.671	9.2	5.65	1100*
ZnTe	(*d*/*Z*)	2.25	530	100	0.18	0.65	6.101	10.4	5.51	1238*
CdS	(*d*/*W*, *Z*)	2.42	250	15	0.21	0.80	4.137	8.9	4.82	1475
CdSe	(*d*/*W*)	1.73	800	—	0.13	0.45	4.30	10.2	5.81	1258
CdTe	(*d*/*Z*)	1.58	1050	100	0.10	0.37	6.482	10.2	6.20	1098
PbS	(*i*/*H*)	0.37	575	200	0.22	0.29	5.936	17.0	7.6	1119
PbSe	(*i*/*H*)	0.27	1500	1500	—	—	6.147	23.6	8.73	1081
PbTe	(*i*/*H*)	0.29	6000	4000	0.17	0.20	6.452	30	8.16	925

All values at 300 K. *Vaporizes

Figure 4.2: Appendix III

$$E_g = 2.42eV.$$

The corresponding wavelength is

$$\begin{aligned}\lambda &= \frac{hc}{E_g} \\ &= \frac{1240eV \cdot nm}{2.42eV} \\ &= 512nm.\end{aligned}$$

From Fig. 4.1, the wavelength corresponding to an absorption coefficient of $\alpha = 1000\ cm^{-1}$ is

$$\lambda_{\alpha=1000cm^{-1}} = 560nm.$$

(B). **10 points in total.**

From Fig. 4.1, the absorption coefficient of Si at a wavelength of 800 nm is

$$\alpha = 850cm^{-1}.$$

Accounting for 30% reflection, the remaining incident optical power is

$$\begin{aligned}P_{\text{rem}} &= 1mW \times 0.70 \\ &= 0.7mW.\end{aligned}$$

Using the Beer-Lambert law, the transmitted optical power is

$$\begin{aligned}P_{\text{out}} &= P_{\text{rem}} e^{-\alpha L} \\ &= 0.7e^{-850\times 4\times 10^{-4}} \\ &= 0.498mW.\end{aligned}$$

The absorbed optical power is

$$\begin{aligned}P_{\text{abs}} &= P_{\text{rem}} - P_{\text{out}} \\ &= 0.7 - 0.498 \\ &= 0.202mW.\end{aligned}$$

The energy of a single photon at $\lambda = 800$ nm is

$$\begin{aligned} E_{\text{ph}} &= \frac{hc}{\lambda} \\ &= 2.48 \times 10^{-19} J. \end{aligned}$$

Assuming 100% quantum efficiency, each absorbed photon generates one electron-hole pair. Therefore,

$$\begin{aligned} N_{\text{EHP}} &= \frac{P_{\text{abs}}}{E_{\text{ph}}} \\ &= \frac{0.202 \times 10^{-3}}{2.48 \times 10^{-19}} \\ &= 8.15 \times 10^{14} s^{-1}. \end{aligned}$$

(C). **4 points in total.**

The sample volume is

$$\begin{aligned} V &= L \times W \times t \\ &= 500 \times 250 \times 4 \mu m^3 \\ &= 5 \times 10^{-7} cm^3. \end{aligned}$$

Assuming that the generated EHPs are uniformly distributed throughout the sample, the optical generation rate is

$$\begin{aligned} g_{\text{op}} &= \frac{N_{\text{EHP}}}{V} \\ &= \frac{8.15 \times 10^{14}}{5 \times 10^{-7}} \\ &= 1.63 \times 10^{21} cm^{-3} s^{-1}. \end{aligned}$$